

Creating Microservices for Account and Loan

In this exercise, I created two Spring Boot microservices named **Account** and **Loan**. Both services provide REST APIs that return sample account and loan details in JSON format. Since both applications initially used the same port, a port conflict occurred. After changing the Loan microservice to use port **8081**, both applications ran successfully and produced the expected output.

Source Code

AccountController.java

```
package com.cognizant.account.controller;
import java.util.HashMap;
import java.util.Map;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RestController;
@RestController
public class AccountController {
    @GetMapping("/accounts/{number}")
    public Map<String, Object> getAccount(@PathVariable String number) {
        Map<String, Object> account = new HashMap<>();
        account.put("number", number);
        account.put("type", "savings");
        account.put("balance", 234343);
        return account;
    }
}
```

LoanController.java

```
package com.cognizant.loan.controller;
import java.util.HashMap;
import java.util.Map;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
```

```

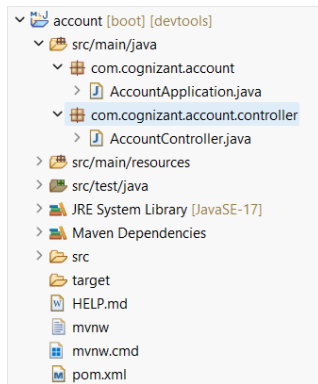
import org.springframework.web.bind.annotation.RestController;
@RestController
public class LoanController {
    @GetMapping("/loans/{number}")
    public Map<String, Object> getLoan(@PathVariable String number) {
        Map<String, Object> loan = new HashMap<>();
        loan.put("number", number);
        loan.put("type", "car");
        loan.put("loan", 400000);
        loan.put("emi", 3258);
        loan.put("tenure", 18);
        return loan;
    }
}

```

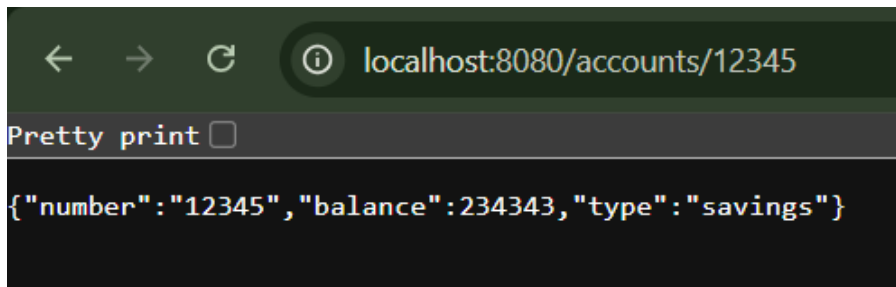
application.properties (Loan)

server.port=8081

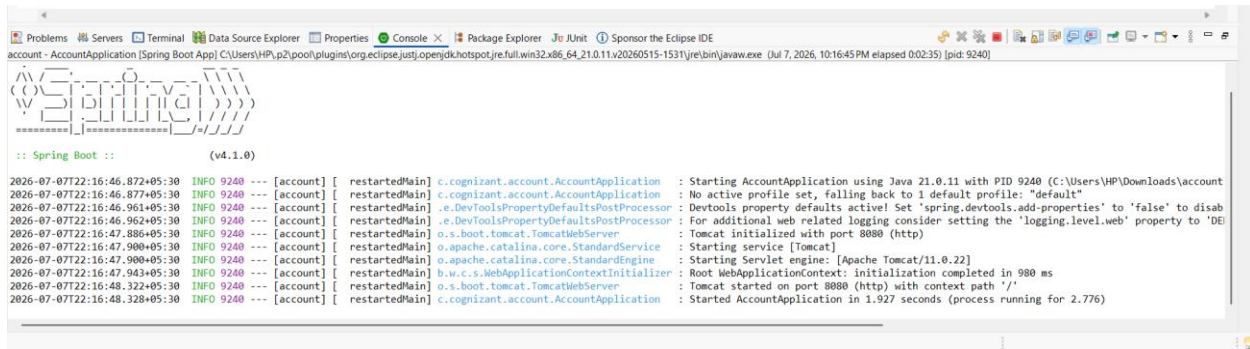
Account Project Structure



Account Browser Output



Account Console Output



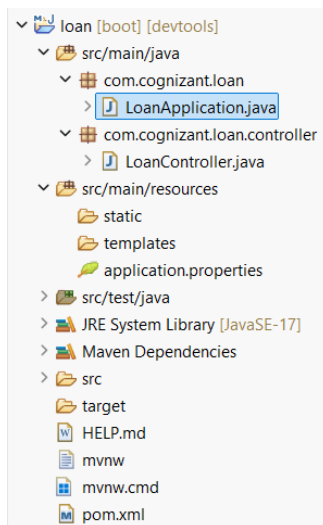
```

:: Spring Boot :: (v4.1.0)

2026-07-07T22:16:46.872+05:30 INFO 9240 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : Starting AccountApplication using Java 21.0.11 with PID 9240 (C:\Users\HP\Downloads\account
2026-07-07T22:16:46.877+05:30 INFO 9240 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-07T22:16:46.961+05:30 INFO 9240 --- [account] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disab
2026-07-07T22:16:46.962+05:30 INFO 9240 --- [account] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property to 'DE
2026-07-07T22:16:47.886+05:30 INFO 9240 --- [account] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-07T22:16:47.900+05:30 INFO 9240 --- [account] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-07T22:16:47.900+05:30 INFO 9240 --- [account] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-07T22:16:47.943+05:30 INFO 9240 --- [account] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 980 ms
2026-07-07T22:16:48.322+05:30 INFO 9240 --- [account] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-07-07T22:16:48.328+05:30 INFO 9240 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : Started AccountApplication in 1.927 seconds (process running for 2.776)

```

Loan Project Structure



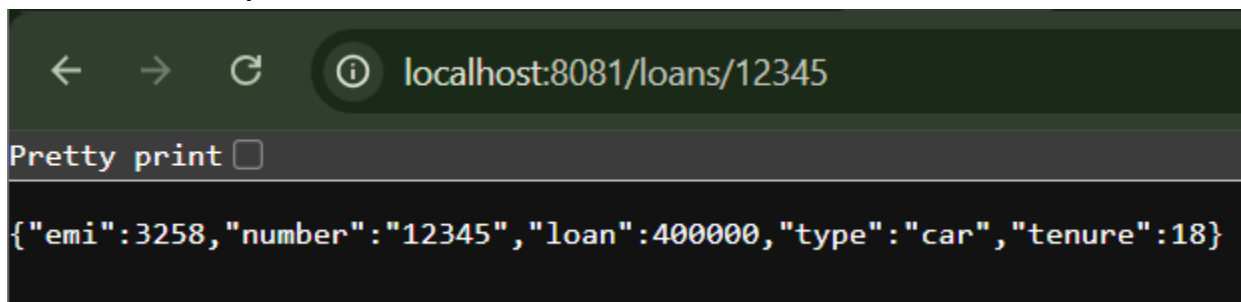
application.properties (Loan)

```

1 spring.application.name=loan
2 server.port=8081

```

Loan Browser Output



Loan Console Output



```
loan - LoanApplication [Spring Boot App] C:\Users\HP\p2\pool\plugins\org.eclipse.jst.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\java.exe (Jul 7, 2026, 10:47:24 PM elapsed 0:00:53) [pid: 27564]

:: Spring Boot :: (v4.1.0)

2026-07-07T22:47:25.727+05:30 INFO 27564 --- [loan] [ restartedMain] com.cognizant.loan.LoanApplication : Starting LoanApplication using Java 21.0.11 with PID 27564 (C:\Users\HP\Downloads\loan\loan\t
2026-07-07T22:47:25.730+05:30 INFO 27564 --- [loan] [ restartedMain] com.cognizant.loan.LoanApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-07T22:47:25.781+05:30 INFO 27564 --- [loan] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
2026-07-07T22:47:25.781+05:30 INFO 27564 --- [loan] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property to 'DEBUG'
2026-07-07T22:47:26.470+05:30 INFO 27564 --- [loan] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8081 (http)
2026-07-07T22:47:26.483+05:30 INFO 27564 --- [loan] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-07T22:47:26.484+05:30 INFO 27564 --- [loan] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-07T22:47:26.530+05:30 INFO 27564 --- [loan] [ restartedMain] o.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 748 ms
2026-07-07T22:47:26.861+05:30 INFO 27564 --- [loan] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8081 (http) with context path '/'
2026-07-07T22:47:26.868+05:30 INFO 27564 --- [loan] [ restartedMain] com.cognizant.loan.LoanApplication : Started LoanApplication in 1.475 seconds (process running for 2.252)
2026-07-07T22:47:38.644+05:30 INFO 27564 --- [loan] [nio-8081-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-07T22:47:38.645+05:30 INFO 27564 --- [loan] [nio-8081-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-07T22:47:38.649+05:30 INFO 27564 --- [loan] [nio-8081-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 3 ms
```

Result

The Account and Loan microservices were created successfully using Spring Boot. Both services returned the expected JSON responses through their REST APIs. The port conflict was resolved by configuring the Loan microservice to use port **8081**, and both applications ran successfully without any errors.